

YIFAN WANG

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SUMMARY

MSc Statistics student at Imperial College London with experience in large-scale user-behaviour data processing, recommender-system evaluation, and spatio-temporal analysis. Processed 21.45M click events from YOOCHOOSE, designed temporal splitting and session cleaning, and compared GRU/LSTM/Bi-GRU models using Recall/MRR/NDCG. Skilled in Python, TensorFlow/Keras, pandas, reproducible experimentation, and results interpretation; seeking data analyst and data science roles.

EDUCATION

MSc Statistics (Data Science & Machine Learning), Imperial College London, UK 09/25 - 11/26

BSc (Hons) Mathematics, University of Liverpool, UK 09/23 - 06/25

BSc Applied Mathematics, Xi'an Jiaotong-Liverpool University, China 09/21 - 06/23

SKILLS

Programming & Tools Python (NumPy, pandas, PyTorch, TensorFlow/Keras, JAX), SQL, Git, LaTeX.

Data & Experiments Large-scale data cleaning, user-behaviour sequence analysis, temporal splitting, feature engineering, batch training, visualisation, and results interpretation.

Modelling & Evaluation Statistical inference, time-series forecasting, recommender systems, Recall/MRR/NDCG, residual-load analysis, and uncertainty evaluation.

Engineering Git, experimental scripting, reproducible results, and technical documentation.

Languages Chinese; English (IELTS 7.0; proficient working language).

PROJECTS

Session-Based Recommender System (YOOCHOOSE) 03/26 - 04/26

Imperial College London

UK

Large-scale clickstream processing, next-item prediction, and ranking evaluation

- Processed 21.45M click events, 6.01M sessions, and 49,499 items; built the data pipeline for timestamp parsing, session cleaning, item-ID remapping, sequence construction, padding, and batching.
- Created a temporal train/test split using the final 24 hours as the test window, preventing sessions from crossing partitions; filtered single-click and over-length sessions and removed test items unseen in training, retaining 16.60M training clicks and 33.4K test clicks.
- Compared Embedding-GRU, LSTM, and Bi-GRU models in TensorFlow/Keras across learning rate, dropout, representation dimension, and batch size; the unidirectional GRU baseline achieved Recall@5=0.326, while Bi-GRU reached 0.741 offline, prompting analysis of possible future-context leakage.

Hillstrom Email Marketing Experiment: A/B Testing and Causal Inference [Project Link](#) 2026.07 – 2026.08

Imperial College London

UK

Treatment-effect estimation, and causal inference for a large-scale email marketing experiment

- Used two-proportion z -tests and Welch's t -tests to estimate pairwise treatment effects; all 9 tests remained significant after Benjamini–Hochberg correction, with the Men's Email yielding the largest lift (visit +7.66pp, conversion +0.68pp, spend +\$0.77 per customer), equivalent to 766 extra visits and \$7,698 extra spend per 10,000 customers, with a 99.9th-percentile Winsorize sensitivity analysis on spend.
- Conducted minimum-detectable-effect analysis at 80% power (visit MDE $\approx$ 0.95pp; spend MDE reduced by 19–21% after Winsorization) and tested 30 treatment-by-segment interactions across 6 dimensions with HC3 robust standard errors and BH adjustment (smallest adjusted $p = 0.406$), favoring broad rollout over segment-specific targeting.
- Implemented single- and multi-covariate CUPED variance reduction ($< 1\%$ on all outcomes, saving 25–374 users) and compared propensity-score matching, IPW, and doubly robust estimation on an artificially selection-biased sample, reducing the visit-effect bias from +14.4% to -1.3% , validating causal-inference corrections for selection bias.

GTFlow: Probabilistic Time-Series Forecasting [Website](#)

04/26 - 08/26

Imperial College London

UK

Uncertainty modelling and forecast evaluation on public time-series datasets

- Addressed TSFlow's under-dispersion from learning with a single observed future by adding Gaussian Process posterior information and window-level trust weighting.
- Evaluated CRPS and Energy Score on 7 GluonTS datasets with 3 random seeds; GTFlow reduced CRPS by 9.3%/13.0% and joint-trajectory Energy Score by 24.3%/17.6% on Solar/Taxi.
- Designed an L1 dispersion-ratio pre-training diagnostic for go/skip dataset decisions; all 21 seed-level decisions fell on the correct threshold side.

EXPERIENCE

Visiting Research Student

07/26 - 09/26

Westlake University

China

- Building a Qwen2.5-0.5B post-training workflow with mlx-lm-lora for causal reasoning and interpretability in LLMs; responsible for SFT, GRPO, and DAPO experiments, log organisation, and result analysis (ongoing).
- In the DAMO project, fine-tuning Qwen2.5-0.5B with SFT, GRPO, and DAPO using mlx-lm-lora; reproduced experiments related to [arXiv:2602.06337](https://arxiv.org/abs/2602.06337) and analysed how causal-inference data and post-training affect causal question answering (ongoing).

AWARDS & SCHOLARSHIPS

- **University of Liverpool:** First Class Honours (Top 2 in cohort), Dean's List, Sampford Memorial, and Willis & Brownlow Hill Prizes.

- **Xi'an Jiaotong-Liverpool University:** Top 50 Excellence Award; Academic Achievement Award (Top 10%).